Shraman Pal

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Education

Purdue University West Lafayette, IN
Ph.D. Candidate; Research: Machine Learning and Operations Research; GPA: 3.93/4.0 *Aug. 2024 - Present*
Centennial Fellowship, 2024 - awarded for outstanding academic merit.

Indian Institute of Technology Kharagpur Kharagpur, India
Integrated M.Tech. in Engineering; Microspecialization in Artificial Intelligence *Jul. 2019 - May 2024*

Technical Skills

Programming and ML Frameworks: Python, Julia, C++, PyTorch, JAX, TensorFlow, Keras
Machine Learning and Optimization: LLMs, Transformers, self-supervised and contrastive learning, constrained reinforcement learning, Bayesian modeling, mixed-integer programming, Bayesian optimization
Research Software: Git, Gymnasium, OmniSafe, Gurobi, AWS, TensorBoard

Experience

Learning Neural Predictions with Explicit Logical Constraints ICML 2026 | GitHub | Paper
Purdue University | Graduate Research Assistant | West Lafayette, IN Aug. 2024 - Present

- Developed an end-to-end neural-symbolic framework that embeds input-dependent logical rules into neural networks, guaranteeing constraint satisfaction on feasible cases; improved low-data classification F1 by **42% relative** (0.30 to 0.43).
- Built and released a reusable Julia package that turns logical constraints into differentiable optimization layers, enabling neural networks to learn through constrained decisions using DiffOpt.jl and Flux.jl.

Building RL Environments and Benchmarks for Agent Evaluation TMLR 2026 | GitHub | Paper
Purdue University | Graduate Research Assistant | West Lafayette, IN Aug. 2024 - Present

- Co-developed an open-source Python benchmark with **9 Gymnasium-compatible constrained-RL environments** spanning planning, scheduling, supply chains, and energy systems.
- Built configurable training and evaluation workflows with OmniSafe integration, parallel environments, CPU/GPU execution, seeded experiments, and Weights & Biases/TensorBoard logging.

Bayesian Dynamics and Predictive Uncertainty ESCAPE 34 / PSE 2024 | Paper
Imperial College London | Machine Learning Research Intern | London, UK May 2023 - Aug. 2023

- Developed a Bayesian hybrid model combining mechanistic equations with learned process dynamics, trained using variational inference and adjoint sensitivities.
- Achieved **3.23-10.61% validation MAPE** for predictions across three process variables; two-standard-deviation predictive intervals covered all held-out observations.

Improving Retrieval-Augmented Question Answering with Data Generation and Quality Filtering Paper
University of Maryland | Machine Learning Research Intern | College Park, MD Jun. 2021 - Dec. 2021

- Built an automated data-generation pipeline that transformed long trivia questions into **770,000+ natural-style QA examples**, using syntactic parsing and a learned BERT-based evaluator to filter malformed questions.
- Augmented RAG training data with quality-filtered, automatically generated questions, improving answer accuracy by **2.8 percentage points** in low-resource experiments.

Learning Aligned Representations from Video and Electrophysiological Signals
University of British Columbia | Machine Learning Research Intern | Vancouver, Canada May 2022 - May 2023

- Designed a video transformer with masked self-supervised pretraining and contrastive alignment to model relationships between behavioral-imaging videos and electrophysiological signals.

Efficient Distributed Training with Shared Gradients AAAI 2022 FL Workshop | Paper
Deakin University | Machine Learning Research Intern | Remote Apr. 2021 - Nov. 2021

- Developed a scalable distributed-learning method that improved FMNIST accuracy by **2.60 points** to **87.48%** across 20 clients.
- Achieved **88.6% lower communication** than SplitFed and **94.7% lower per-client computation** than federated learning, while remaining within **0.4 percentage points** of federated-learning accuracy in the resource-comparison experiment.

Building Multimodal Models for Image Classification and Text-to-Image Retrieval
ChiSquareX | Machine Learning Engineer | Kharagpur, India Nov. 2020 - Jan. 2022

- Developed multimodal deep-learning models and training pipelines for image classification and text-to-image retrieval.

Additional Research

Surrogate Modeling and Bayesian Design Optimization Digital Chemical Engineering, 2024 | Paper

- Built LightGBM surrogate models for catalyst performance, achieving $R^2 = 0.94$ for CO conversion and $R^2 = 0.88$ for temperature; the study used Bayesian optimization to search catalyst compositions while balancing activity, stability, and cost.